

MANGU 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

233/3

CHEMISTRY

PAPER 3 (PRACTICAL)

TIME: 2¼ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

Instructions:

- Answer **ALL** questions in the spaces provided.
- You are **NOT** allowed to start working with the apparatus for the first 15 minutes of the 2 ¼ hours allowed for this paper. This time will enable you read through the question paper and make sure you have all the chemicals and apparatus required.
- Mathematical tables and electronic calculators may be used
- All working must be clearly shown where necessary.

FOR EXAMINER'S USE ONLY.

Question	Maximum score	Candidate's score
1	20	
2	12	
3	8	
Total score	40	

1. You are provided with:

- Solution A of Potassium manganate (VII).
- 0.05M solution B of oxalic acid.
- Solution C containing 4.9g of ammonium iron (II) Sulphate, $(\text{NH}_4)_2 \text{SO}_4 \cdot \text{FeSO}_4 \cdot 6\text{H}_2\text{O}$, in 250cm^3 of water.
- You are required to:
 - i) Determine the rate of reaction between oxalic acid and Potassium manganate (VII) solutions.
 - ii) Standardize the solution A.

PROCEDURE I:

- (i) Fill the burette with solution A.
- (ii) Place 1 cm^3 of solution A from the burette into each of the five (5) test-tubes in a test tube rack.
- (iii) Using a clean measuring cylinder, place 19 cm^3 of solution B into a boiling tube.
- (iv) Place the thermometer into solution B and heat gently until it attains a temperature of 40°C .
- (v) Add the first portion of solution A **immediately and at the same time start a stop watch**.
- (vi) Record the time taken for solution A to be decolorized in table I below.
- (vii) Repeat the procedure (i) to (v) at temperatures of 50°C , 60°C , 70°C and 80°C to complete the table.

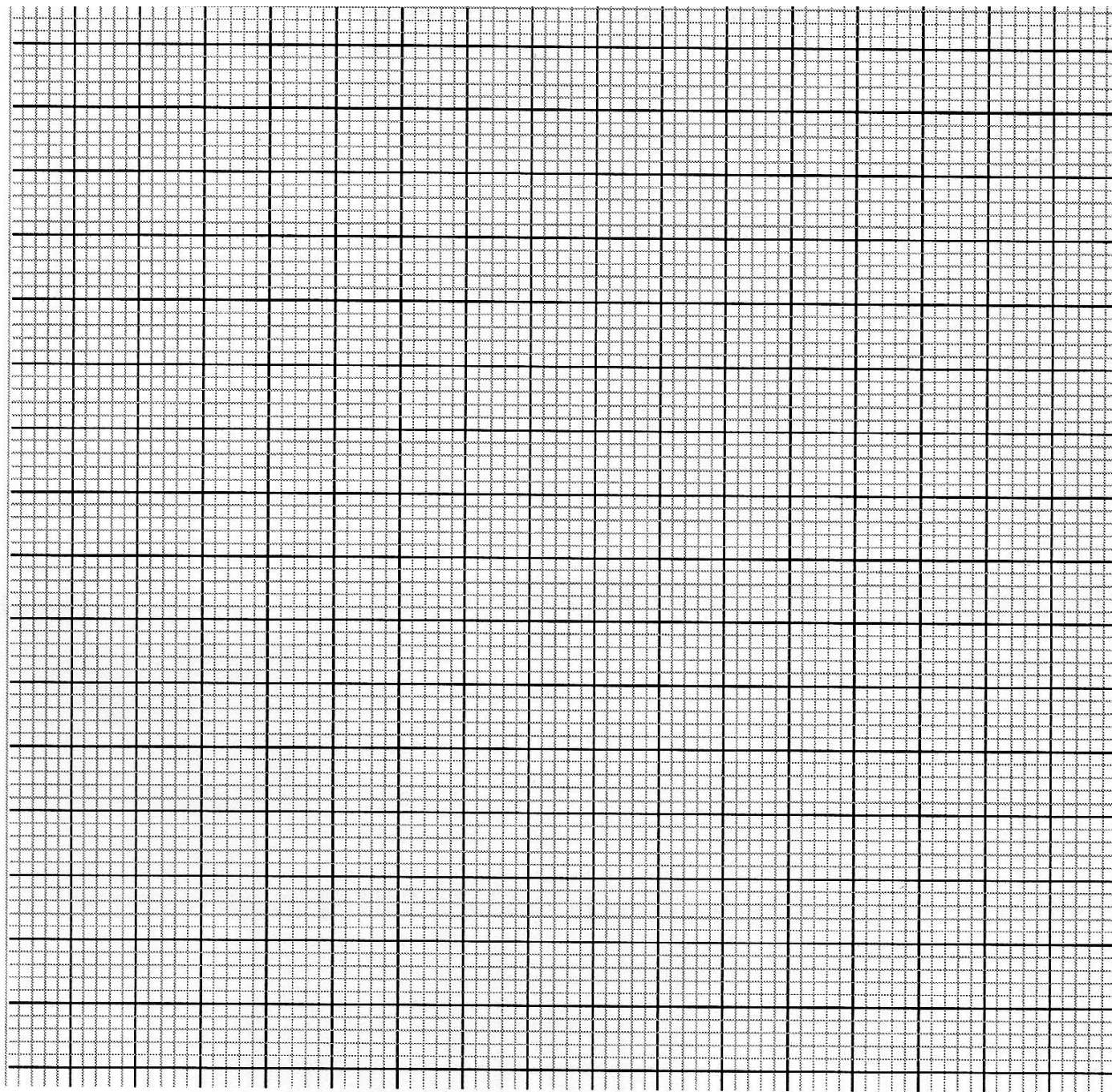
Table I

Temperature of solution B ($^\circ\text{C}$)	40	50	60	70	80
Time taken for decolorization (seconds)					
Rate $(\frac{1}{t})\text{ s}^{-1} \times 1000$					

(4 marks)

i) Plot a graph of ($\frac{1}{t} \times 1000$) against temperature (X-axis)

(3 marks)



ii) From the graph determine the time taken for the mixture to decolourise at 65°C **(2 marks)**

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- iii) How does the rate of reaction between oxalic acid solution B and Potassium manganate (VII) solution A vary with temperature? (1 mark)
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PROCEDURE II

- Refill the burette with solution A.
- Pipette 25cm³ of solution C into a conical flask and titrate the solution A against solution C until a permanent pink colour just appears.
- Record your results in table II below and repeat the procedure to fill the table.

Table II	1	2	3
Final burette reading (cm ³)			
Initial burette reading (cm ³)			
Volume of A used (cm ³)			

(4 marks)

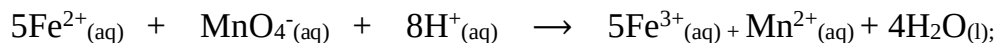
- i) Determine the average volume of A used. (1 mark)
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- ii) Calculate the concentration of solution C in moles per litre (Fe = 56, S = 32, O = 16, N = 14, H = 1) (1 mark)
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- iii) Find the number of moles of solution C used. (1 mark)

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 iv) Given the ionic equation for the reaction is



Find the number of moles of solution A used.

(1 mark)

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 (v) Determine the concentration of the Potassium manganate (VII), solution A in moles per litre.

(2 marks)

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 2. You are provided with solid Q. Carry out the tests below and record your observations and inferences in the table below.

i) Place half a Spatula full of solid Q in a clean dry test-tube and heat gently then Strongly.

Test the gas produced using moist red and blue litmus papers

Observations	Inferences
(1 mark)	(1 mark)

ii) Place the remaining solid Q in a boiling tube, add about 5cm³ of distilled water and shake well. Divide the resulting mixture into four portions for the tests below.

Observations	Inferences
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(1 mark)	(1 mark)
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a). To the first portion add Sodium hydroxide solution dropwise until in excess.

Observations	Inferences
(1 mark)	(1 mark)

b). To the second portion add 2-3 drops of dilute Sulphuric (VI) acid

Observations	Inferences
(1 mark)	(1 mark)

c). To the third portion add aqueous ammonia dropwise until in excess

Observations	Inferences
(1 mark)	(1 mark)

d). To the fourth portion add 2-3 drops acidified barium nitrate solution

Observations	Inferences
(1 mark)	(1 mark)

3. You are provided with solid L. carry out the tests below on L and record the observations and inferences in the spaces provided.

(a) Place **half** of solid L in a boiling tube and add about 5cm³ of distilled water

(i)

Observations	Inferences
(1 mark)	(1 mark)

Divide the solution into two portions and carry out the tests below.

(ii). To the first portion add 2-3 drops of acidified potassium manganate (VII).

Observations	Inferences
(1 mark)	(1 mark)

(iii). To the second portion add Sodium carbonate provided.

Observations	Inferences
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(1 mark)	(1 mark)
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(b). Place the remaining solid L in metallic spatula and ignite it.

Observations	Inferences
(1 mark)	(1 mark)

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